

Homework 3

Due: Wednesday 06/05/2026 11:00AM

Nộp theo nhóm

Submission: L0X_HW2_YOURNAME.zip (X = 1, 2)

Accepted file: ipynb + pdf

Genetic Algorithm (GA)

Choose ONE of the 3 topics below:

Your GROUP report should include source code and a detailed documentation of the algorithms.

You must design **Fitness function, Selection, Crossover, Mutation**.

The datasets for Topic1 and Topic 2 are included.

Topic 1 (Max 8/10):

Traveling Salesman Problem (TSP) using Genetic Algorithm.

Number of cities: 51 (The file *TSP51_dataset.txt* shows the city number and its x, y coordinates).

Assume that all cities are connected to one another **EXCEPT** city 1 and city 22, city 1 and city 32 (There is NO direct path from city 1 to city 22 or from city 1 to city 32).

Plot the final path of the Traveling Salesman.

Also solve the problem again with **Simulated Annealing** and compare it with GA algorithm

Topic 2 (Max 9/10):

Job shop scheduling using Genetic Algorithm

This is a 10x10 Job shop problem with 10 workloads and 10 machines. Each workload is processed in different order of machines. (*JobShop_dataset.xlsx*)

The sequence of processing operations is presented. Each workpiece will go through 10 processing operations but the order of machines is different. For example: Job-02 uses Machine 1 (for 43m), then Machine 3 (90m), Machine 5 (75m), Job-03 uses Machine 2 (91m), then Machine 1 (85m), Machine 4 (39m), ...

The scheduling goal is to minimize the total completion time or Makespan.

Note: Makespan is the total time elapsed from the start to the completion of a set of jobs, representing the maximum completion time among all machines or resources.

Show your final output and makespan plot.

Processing Time (minutes)

Time ▾	T1 ▾	T2 ▾	T3 ▾	T4 ▾	T5 ▾	T6 ▾	T7 ▾	T8 ▾	T9 ▾	T10 ▾
Job-01	29	78	9	36	49	11	62	56	44	21
Job-02	43	90	75	11	69	28	46	46	72	30
Job-03	91	85	39	74	90	10	10	89	45	33
Job-04	81	95	71	99	9	52	85	98	22	43
Job-05	14	6	22	61	26	69	21	49	72	53
Job-06	84	2	52	95	48	72	47	65	6	25
Job-07	46	37	61	13	32	21	32	89	30	55
Job-08	31	86	46	74	32	88	19	48	36	79
Job-09	76	69	76	51	85	11	40	89	26	74
Job-10	85	13	61	7	64	76	47	52	90	45

Machine Sequence (Order)

order ▾	O1 ▾	O2 ▾	O3 ▾	O4 ▾	O5 ▾	O6 ▾	O7 ▾	O8 ▾	O9 ▾	O10 ▾
Job-01	1	2	3	4	5	6	7	8	9	10
Job-02	1	3	5	10	4	2	7	6	8	9
Job-03	2	1	4	3	9	6	8	7	10	5
Job-04	2	3	1	5	7	9	8	4	10	6
Job-05	3	1	2	6	4	5	9	8	10	7
Job-06	3	2	6	4	9	10	1	7	5	8
Job-07	2	1	4	3	7	6	10	9	8	5
Job-08	3	1	2	6	5	7	9	10	8	4
Job-09	1	2	4	6	3	10	7	8	5	9
Job-10	2	1	3	7	9	10	6	4	5	8

Topic 3 (Max 10/10):

Class Scheduling using Genetic Algorithm

num_courses = 15

num_sections = 10

course_name = {1-15}. Note a course can repeat. For example: Course 1 is scheduled on Mon-Wed, section 01, ALSO another course 1 is scheduled Tue-Thu, section 02.

num_professors = 10 (namely 1-10)

num_days = 5 (2-6 or M-T-W-R-F)

num_timeslots_per_day = 6 (timeslot 1, 2, ... ,6 during morning - afternoon)

num_rooms = 30

room_size = {0, 1} (2 types only: 0 for Small rooms, 1 for Large rooms)

class_registration_size = {0, 1} (room size 1 can be used for class_registration of size 0 or 1
BUT room size 0 can ONLY be used for class_registration of size 0)

Constraints

1. A course has 2 sessions (lectures) per week (for example Monday and Thursday), **NOT** on the **same** or **adjacent** days. For example: a course is scheduled on **Monday** and **Tuesday (next day)** is not acceptable. a course scheduled on **Monday** morning and **Monday** afternoon is not acceptable.
2. Course's class registration size and room size should be reasonable. For example: room size 1 can be used for class registration of size 0 or 1, BUT room size 0 can ONLY be used for class registration of size 0.
3. A professor cannot teach two courses at the same time. No professor can teach more than 3 courses.
4. Classes are scheduled only in available classrooms.
5. A section cannot be scheduled in two different rooms simultaneously.
6. A room cannot be assigned to two different sections at the same time.

Generate schedule(s) with minimal conflicts between sections, professors, and rooms. **Show your final class schedule** and the fitness plot.

Note: You might need to initialize some data about classes, rooms, professors, ... with randomization before you could use GA.

--- The end ---